Spatial-TACNA

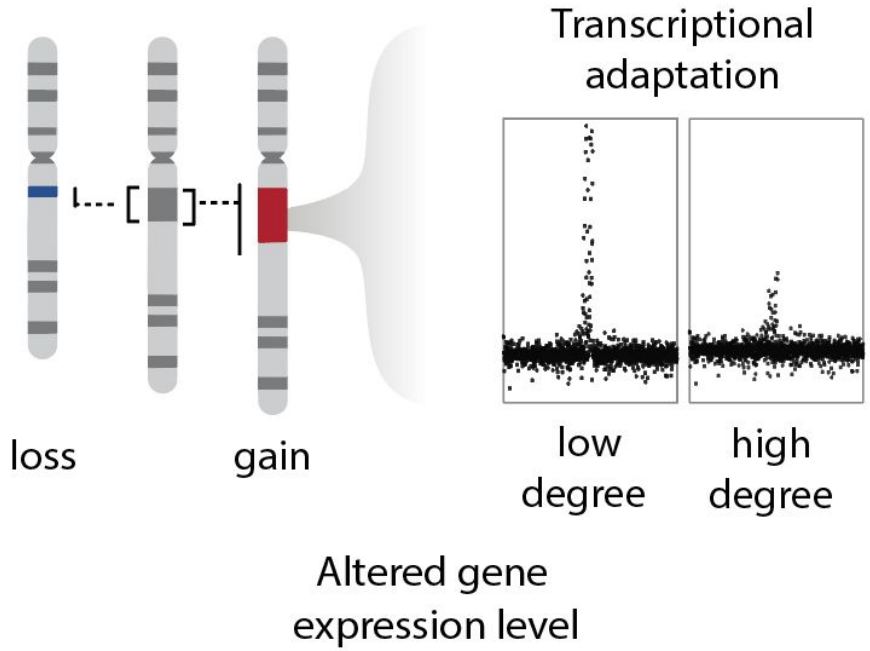
A novel methodology for unravelling the effects of copy number alterations in spatial transcriptomics

Feija Somefun, Marco de Bruyn, Arkajyoti Bhattacharya, Rudolf S.N. Fehrmann

University Medical Center Groningen, The Netherlands

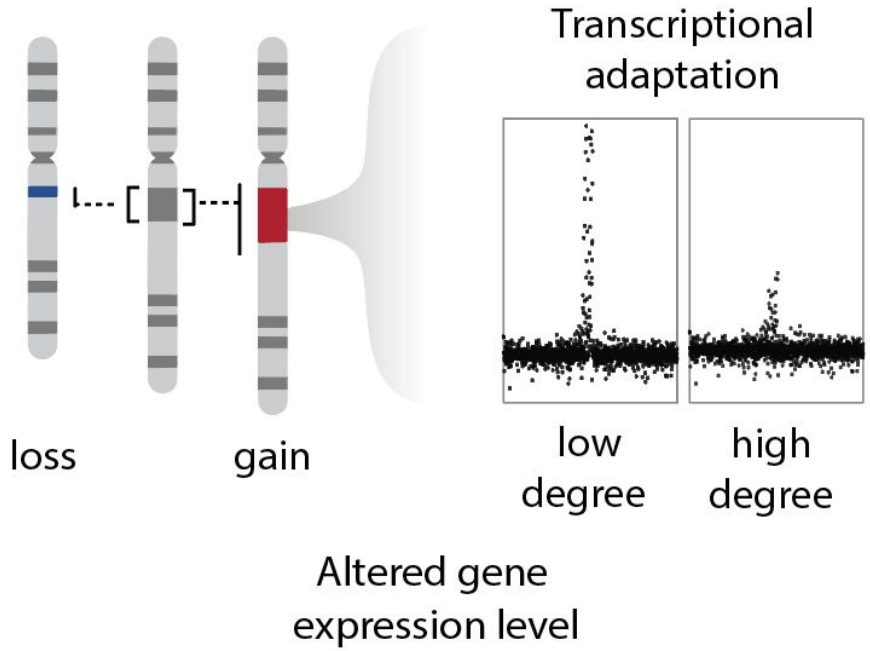
Copy number alterations ....

Copy number alterations (CNAs)

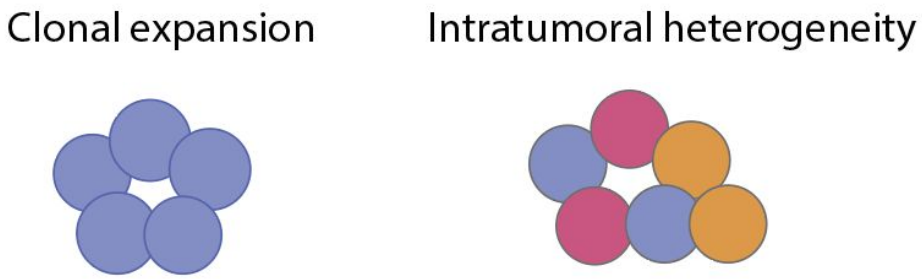


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Copy number alterations (CNAs)

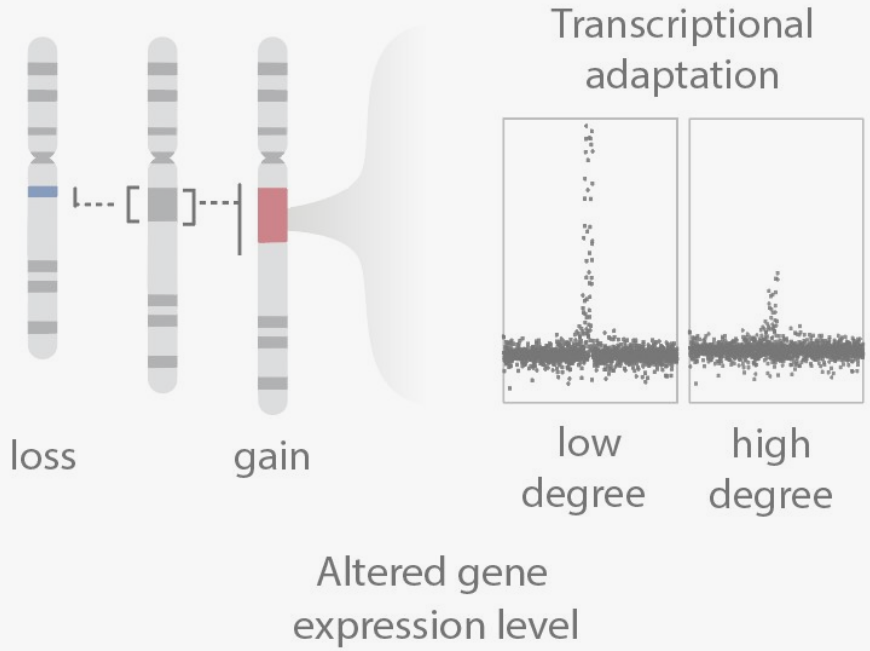


Spatial transcriptomics

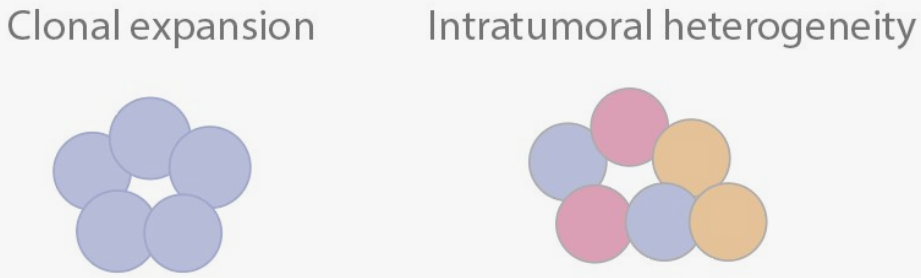


Copy number alterations ....

Copy number alterations (CNAs)



Spatial transcriptomics



Aim: Accurately infer copy number alterations in spatial gene expression profiles /infer CNAs in spatial gene expression profiles while preserving local spatial variation.

Spatial-TACNA infers CNAs in bulk transcriptomics

~10k bulk gene expression profiles

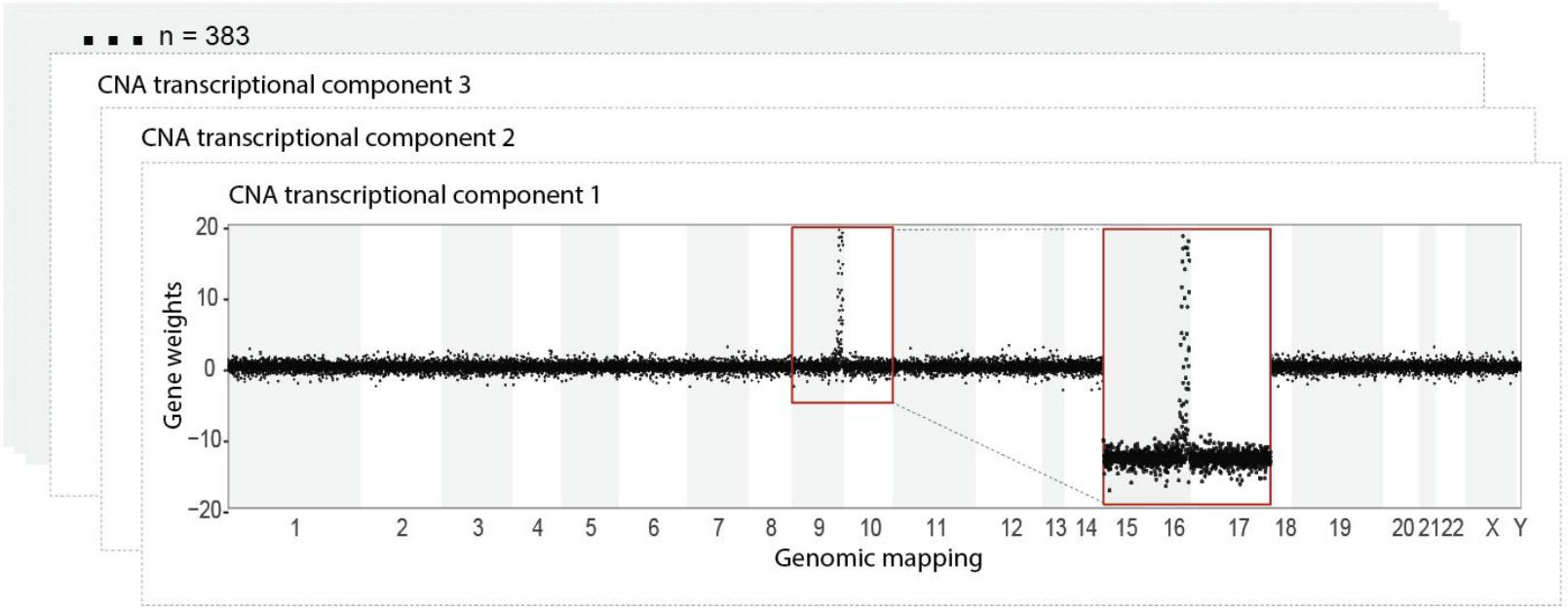


Independent Component Analyses

Spatial-TACNA infers CNAs in bulk transcriptomics

~10k bulk gene expression profiles

Independent Component Analyses



Spatial-TACNA infers CNAs in bulk transcriptomics

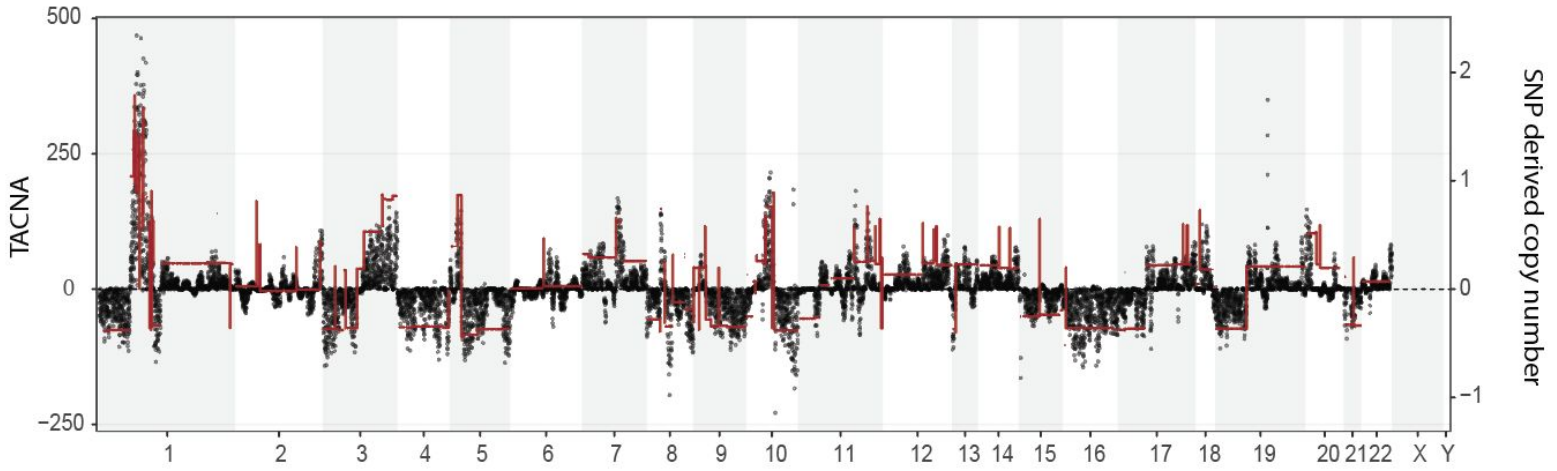
~10k bulk gene expression profiles

Independent Component Analyses

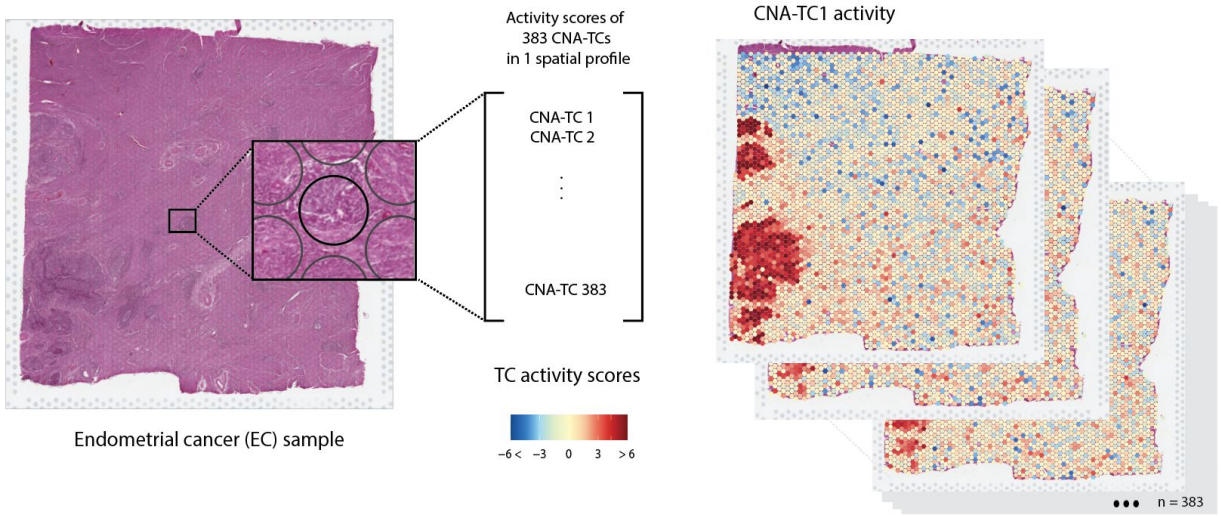
Transcriptional adaptation to copy number profiles (TACNA)

TACNA profile

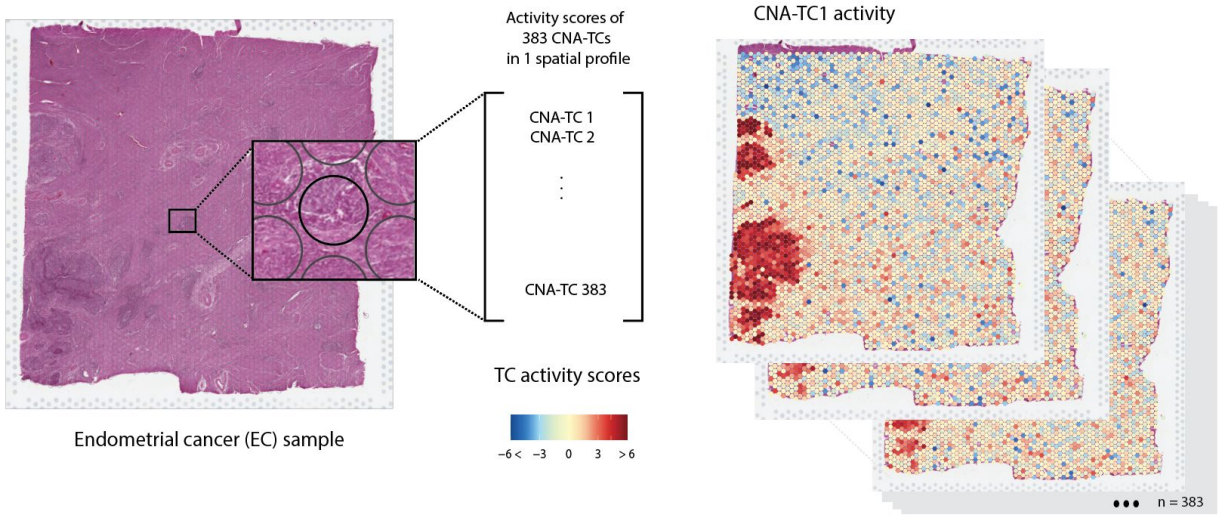
SNP derived CNA profile



Spatial-TACNA infers CNAs in spatial transcriptomics



Spatial-TACNA infers CNAs in spatial transcriptomics



CNA- immune interactions

CNA- functional pathways

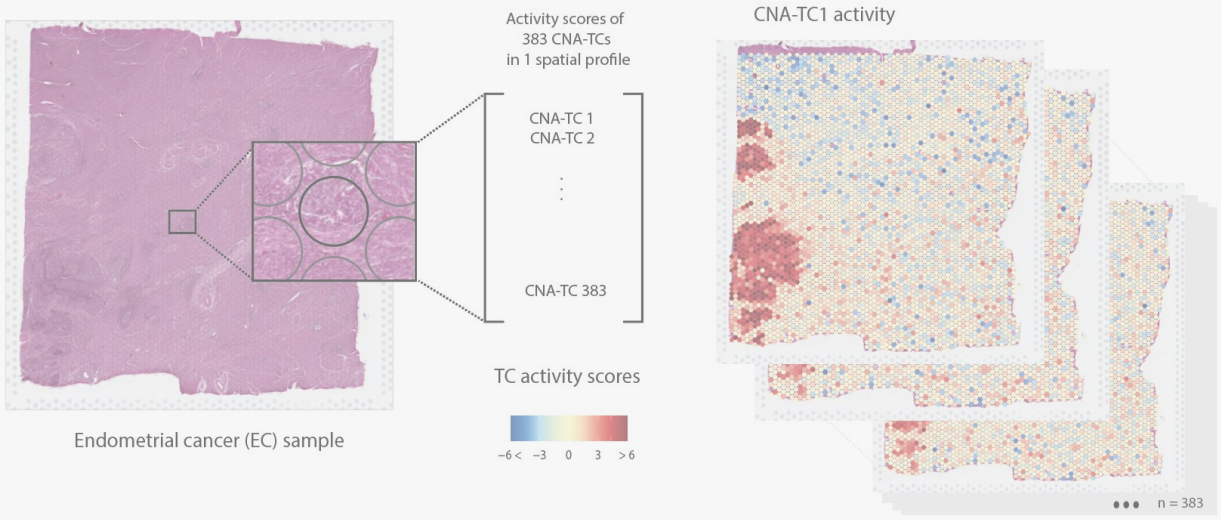
Copy number burden heterogeneity

Copy number burden aligns with histological tumor regions



Spatial-TACNA infers CNAs in spatial transcriptomics

Diverse effects of distinct CNAs identified in spatial gene expression profiles



CNA- immune interactions

CNA- functional pathways

Copy number burden heterogeneity

Copy number burden aligns with histological tumor regions



Spatial-TACNA: provides a framework for investigating effects of CNAs in their spatial context / the spatial heterogeneity and transcriptional consequences of CNAs within tumors

Conclusion: Spatial-TACNA infers CNAs in bulk and spatial transcriptomics samples.

QR code to email and details

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